

PRODUCT SPECIFICATION

Front Door Power Window Control Module (PWCM)

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Author	Body Systems Engineering	Classification	Internal

1. Introduction

1.1 Purpose

This document specifies the functional and non-functional behaviour required of the Front Door Power Window Control Module (hereafter PWCM). The PWCM is the electronic control unit responsible for driving the window regulator motor in a single front door, interpreting occupant switch requests, detecting obstructions during closing movement, and reporting its status to the body domain controller.

This specification is the input to software requirements analysis and to the definition of the qualification test campaign for the module.

1.2 Scope

The scope of this document covers the control logic resident in the PWCM. It does not cover:

- The mechanical design of the window regulator, cable drive, or door seal system.
- The design of the door switch pack hardware, other than its electrical interface to the PWCM.
- Vehicle-level functions such as global open/close from the remote key, which are commanded by the body domain controller and consumed by the PWCM as ordinary network requests.

Where behaviour depends on a vehicle-level function, the PWCM is specified only as far as its own response to the incoming request.

1.3 Definitions

Term	Meaning
PWCM	Front Door Power Window Control Module, the subject of this specification.
BDC	Body Domain Controller. Supervisory ECU that issues vehicle-level window commands.
Express movement	A movement that continues to the end stop after the switch has been released.
Manual movement	A movement that continues only while the switch is held.
Reversal	The safety response in which a closing window halts and re-opens after an obstruction is detected.
Normalisation	The learning cycle by which the module establishes the physical end-stop positions of the glass.

2. System Overview

The PWCM sits between the door switch pack and the window regulator motor. It receives discrete switch inputs from the occupant, receives supervisory commands from the BDC over the body CAN network, and drives the motor through an integrated H-bridge output stage. Motor rotation is monitored by a Hall-effect sensor pair integrated into the motor assembly, from which the module derives both glass position and instantaneous motor load.

The module maintains an internal model of glass position expressed as a percentage of the closed position, where 0 % represents fully open and 100 % represents fully closed against the upper seal. Because the module retains its learned end-stop positions and last known glass position across ignition cycles, a routine restart does not require the occupant to re-normalise the window before express movements are again available.

The module is powered from the door harness and remains partially active during bus sleep in order to service local switch inputs for a limited period after ignition off.

3. Operating Modes

3.1 Manual movement

A short actuation of the window switch to the first detent commands manual movement. The window travels in the requested direction for as long as the switch is held at the first detent, and stops when the switch is released or when the glass reaches the relevant end stop, whichever occurs first.

The window shall respond quickly to switch actuation so that the occupant does not perceive a delay between operating the switch and the glass beginning to move.

3.2 Express movement

A deeper actuation to the second detent commands express movement. Express movement continues after the switch has been released and runs until the glass reaches the end stop, until an obstruction is detected, or until the occupant interrupts it.

Express down is available on both front doors. Express up is available on the driver's door only, and only while the module holds a valid normalisation record. The occupant interrupts an express movement by actuating the switch again in either direction; the movement stops and no new movement is started by that interrupting actuation.

3.3 Express-up sequence

When express up is commanded the module first confirms that a valid normalisation record is held and that no obstruction fault is currently latched. It then energises the motor in the closing direction and monitors motor load continuously. As the glass approaches the upper seal the module enters the seal engagement region, within which the load rise caused by the seal itself is expected and is therefore not treated as an obstruction. Once the module observes the load signature characteristic of the glass having seated against the seal, it de-energises the motor and records the position as the closed end stop. If the load signature is not observed within the expected travel, the module de-energises the motor, discards the normalisation record, and reports a position plausibility fault; express up is then unavailable until a successful normalisation has been performed by a manual close held to the end stop.

3.4 Rear window lockout (child lock)

The driver's switch pack carries a lockout switch. When lockout is active, window requests originating from any switch pack other than the driver's are ignored. Requests originating from the driver's switch pack and requests arriving from the BDC remain effective while lockout is active. The lockout state is retained across ignition cycles.

4. Interfaces

4.1 Inputs

Signal	Type	Description
SW_UP_D1	Discrete	Driver switch, up, first detent (manual close request).
SW_UP_D2	Discrete	Driver switch, up, second detent (express close request).
SW_DN_D1	Discrete	Driver switch, down, first detent (manual open request).
SW_DN_D2	Discrete	Driver switch, down, second detent (express open request).
SW_LOCKOUT	Discrete	Rear window lockout switch state.
HALL_A / HALL_B	Pulse pair	Quadrature motor rotation signal; source of position and load estimation.
V_BATT	Analogue	Door harness supply voltage.
T_MOD	Analogue	Module substrate temperature.
CAN_WIN_REQ	Network	Window command from the BDC (open, close, stop, express variants).

4.2 Outputs

Signal	Type	Description
MTR_PWM	PWM	Motor drive duty command to the H-bridge.
MTR_DIR	Discrete	Motor direction selection.
CAN_WIN_STAT	Network	Glass position, movement state, and availability of express functions.
CAN_WIN_DTC	Network	Active diagnostic trouble codes.

5. Performance Parameters

The following parameters apply across the full specified operating range unless stated otherwise.

Parameter	Description	Value	Tolerance / Note
P-01	Maximum squeezing force during closing	100 N	Applies in the detection zone defined in §6.1
P-02	Obstruction detection zone, lower bound	4 mm	Measured from the mating vehicle structure
P-03	Obstruction detection zone, upper bound	200 mm	Measured from the mating vehicle structure
P-04	Motion halt time following obstruction detection	100 ms	Maximum
P-05	Reversal travel after obstruction	≥ 300 mm	Or to fully open if less travel remains
P-06	Full travel time, express down	1.4 s	± 0.2 s at 13.5 V, 23 °C
P-07	Full travel time, express up	1.7 s	± 0.2 s at 13.5 V, 23 °C
P-08	Operating voltage range	9.0 – 16.0 V	Full function
P-09	Degraded voltage range	6.5 – 9.0 V	Manual movement only
P-10	Operating temperature range	–40 – +85 °C	Module substrate
P-11	Consecutive full-travel cycles before thermal derate	12	See §7.3
P-12	Local switch service time after ignition off	TBD	Target 10 min, to be confirmed

6. Obstruction Detection and Reversal

6.1 Detection zone

An obstruction present between 4 mm and 200 mm from the vehicle structure against which the glass closes shall be detected during any closing movement that is not under continuous occupant supervision. Within 4 mm of the mating structure the reversal function is suppressed, because the residual gap is too small to admit a body part and because suppression avoids nuisance reversals caused by seal friction, ice, or minor misalignment.

6.2 Detection method

Obstruction detection is based on motor load estimated from the Hall signal. The module continuously compares the instantaneous load against an expected load profile learned during normalisation and corrected for supply voltage and module temperature. A deviation exceeding the calibrated threshold for longer than the debounce interval is treated as an obstruction.

The expected load profile accounts for the seal engagement region described in §3.3, so that the load rise produced by the seal is not itself interpreted as an obstruction.

6.3 Reversal response

On detection the module shall halt the closing movement within 100 ms and shall then drive the glass in the opening direction by at least 300 mm, or to the fully open position where less travel remains. The reversal movement shall not be interruptible by a further close request until it has completed.

The force exerted on an obstruction shall not exceed 100 N at any point during a closing movement within the detection zone.

6.4 Repeated obstruction

If three obstruction events occur within the same closing attempt sequence, the module shall inhibit further automatic closing movement until the ignition has been cycled, and shall report the condition to the BDC. Manual closing under continuous occupant supervision remains available.

7. Error Handling and Degraded Operation

7.1 Loss of position confidence

If the Hall signal is lost, becomes implausible, or if the module detects that the glass has moved while the module was not driving it, the normalisation record is discarded. Express movements become unavailable and the module reports reduced availability. Normalisation is re-established by a manual close movement held until the glass reaches the upper end stop.

7.2 Supply voltage excursions

Below the full-function voltage range the module suspends express movements and permits manual movement only. Below the degraded range the module de-energises the motor and takes no further movement request until supply recovers. Above the full-function range the module de-energises the motor and reports an overvoltage condition.

7.3 Thermal protection

Repeated full-travel cycles heat the motor and the output stage. After the number of consecutive cycles given by P-11, the module progressively reduces available drive until a cool-down period has elapsed. Manual movement remains available throughout derating, at reduced speed.

7.4 Communication loss

If the BDC becomes unreachable, the module continues to service local switch inputs using its last known lockout state. Vehicle-level commands are not actioned while communication is lost. On restoration the module resynchronises its reported status before accepting new vehicle-level commands.

7.5 Fault reporting

Condition	Module response	Reported
Hall signal implausible	Discard normalisation; express unavailable	Yes
Obstruction inhibit active	Automatic closing inhibited	Yes
Overvoltage	Motor de-energised	Yes
Undervoltage, degraded band	Manual movement only	Yes
Thermal derate active	Reduced drive	Yes
Position plausibility fault	Express unavailable; normalisation required	Yes

8. Regulatory Context

The obstruction detection and reversal behaviour specified in §6 is derived from the requirements applicable to power-operated window systems in the target markets. Publicly available regulation in both the United States and under the UNECE framework requires that a power-operated window closing without continuous occupant supervision must reverse before exerting a squeezing force of 100 N on an obstruction located between 4 mm and 200 mm from the structure against which the window closes.

The parameter values P-01 through P-03 are set directly by this regulatory context and shall not be re-calibrated without a formal change to this specification.

Compliance is demonstrated at vehicle level. This specification constrains only the module behaviour that enables that demonstration.